ELEC 205

Assignment 1: Gate-Level Minimization and Design

Due by March 26, 11:59 PM

In the design assignments, you will be using the Logisim logic simulator. Logisim provides a graphical user interface for designing and simulating digital circuits. Logisim includes libraries of basic gates, multiplexers, decoders, memory elements, and other simple components.

Logisim Installation:

To get started, follow these instructions to install Logisim on your working machine.

1. Logisim requires Java 5 or later. Ensure that you have it on your computer. Note that Java is available from java.sun.com.
2. Download Logisim Evolution from the GitHub project. It is available for a wide range of operating systems.

Please use Logisim to answer the questions below. **Handwritten solutions will not be graded.** A very short tutorial is available at https://www.youtube.com/watch?v=cMz7wyY_PxE&ab_channel=BarryBrown. Many other Logisim tutorials are available on YouTube.

It is essential that your submitted report is clear, legible, and easy to understand. Any low-quality or illegible screenshots, tables, or equations will be considered invalid. Please prepare your submission as a proper technical report. The use of AI tools is strictly prohibited; both the work and the report must be entirely your own.

For the solutions, please take screenshots of Logisim and include them in your submission file. If you face any problem in Logisim (for example, if 'F' and 'G' are not the same in *Project → Analyze Circuit → Expression,* even though the circuit is implemented correctly in Question 3 Part (e)), please try to change *Project → Analyze Circuit → Minimized → Format* (SOP or POS) a few times, and the error should be fixed.

Introduction to Basics and Learning Logisim

Question 1

- Simplify and implement the following Boolean function:

$$F(A, B, C, D) = \prod(1, 3, 5, 6, 9, 11, 12, 14)$$

Hint: Fill the truth table from Project → Analyze Circuit → Table. (You may want to place the input and output pins first and label them as A, B, C, D, or F)

Question 2

Implement the following expressions (do not use *Project → Analyze Circuit → Build Circuit)* and show their truth tables and Karnaugh maps. Lastly, simplify them to sum-of-products and products-of-sums. *Hint: Use Project → Analyze Circuit → Minimized → Format.*

(a) $F(A, B, C, D) = ACD' + C'D + AB' + ABCD$

(b) $F(A, B, C, D) = (A' + B + D')(A' + B' + C')(A' + B' + C)(B' + C + D')$

Question 3

Please follow the steps below.

- Implement the following expression using only NAND gates.

$F = BCD + ACD + ABD$

Hint: You can use Project → Analyze Circuit → Expression → Build Circuit. When you click Build Circuit, check the options.

(a) Show the corresponding Karnaugh map

(b) Implement another circuit on the same page by converting all the gates to NOR gates (you canuse additional inverters) and label the output of the new circuit as "*G*".

(c) Observe that the outputs *F* and *G* are the same by changing the inputs *A, B, C,* and *D*.

(d) Check the corresponding Karnaugh maps for *F* and *G* and show them that they are the same.

Design Problem

- Design a combinational circuit that accepts two 2-bit unsigned inputs (AB, CD) and generates a 2-bit unsigned output (XY) equal to the absolute difference between the two inputs. For example, if the two 2-bit inputs are 01 and 00, the output is 01, or if the two 2-bit inputs are 00 and 01, the output is again 01.

Your assignment report should include the following work:

a. A clear presentation of your design, including your truth table, k-map minimization, and minimal combinational output function.
b. Schematic entry of your designed circuit in Logisim. Label all inputs and outputs. Include a screenshot of your schematic entry in your report.
c. Under Project → Analyze Circuit, browse and investigate I/O, Table, Expression, and Minimized tabs. You may represent don't care conditions with the dash symbol "-". Report the correctness of your design by including screenshots.
d. Under Simulate → Timing Diagram, browse and investigate timing diagram analysis by changing the 4-bit inputs of your circuit, and observe outputs. Report the correctness of outputs and include a screenshot of the timing diagram.